



SVM PRESENTATION

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01 - HISTORY

02 - BASIC PRINCIPLE

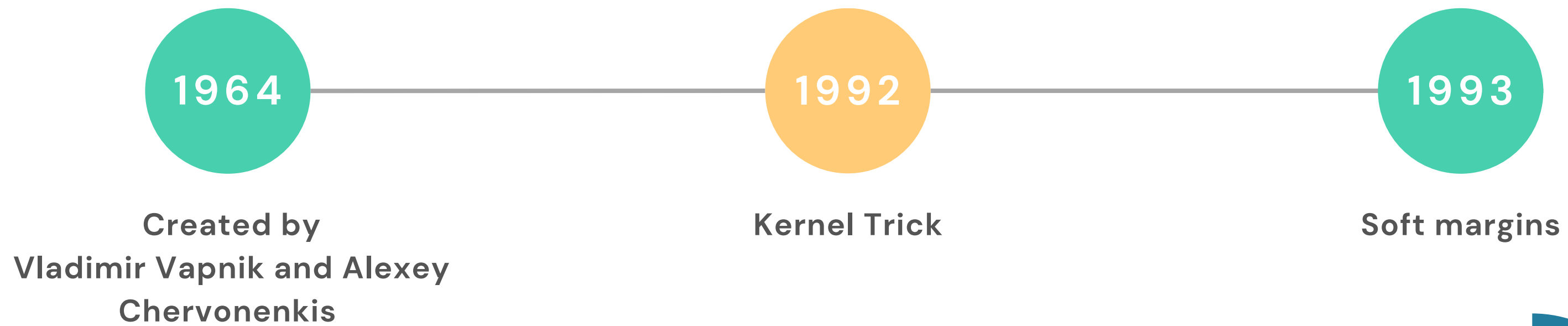
03 - HOW IT WORKS

04 - KEY FORMULAS

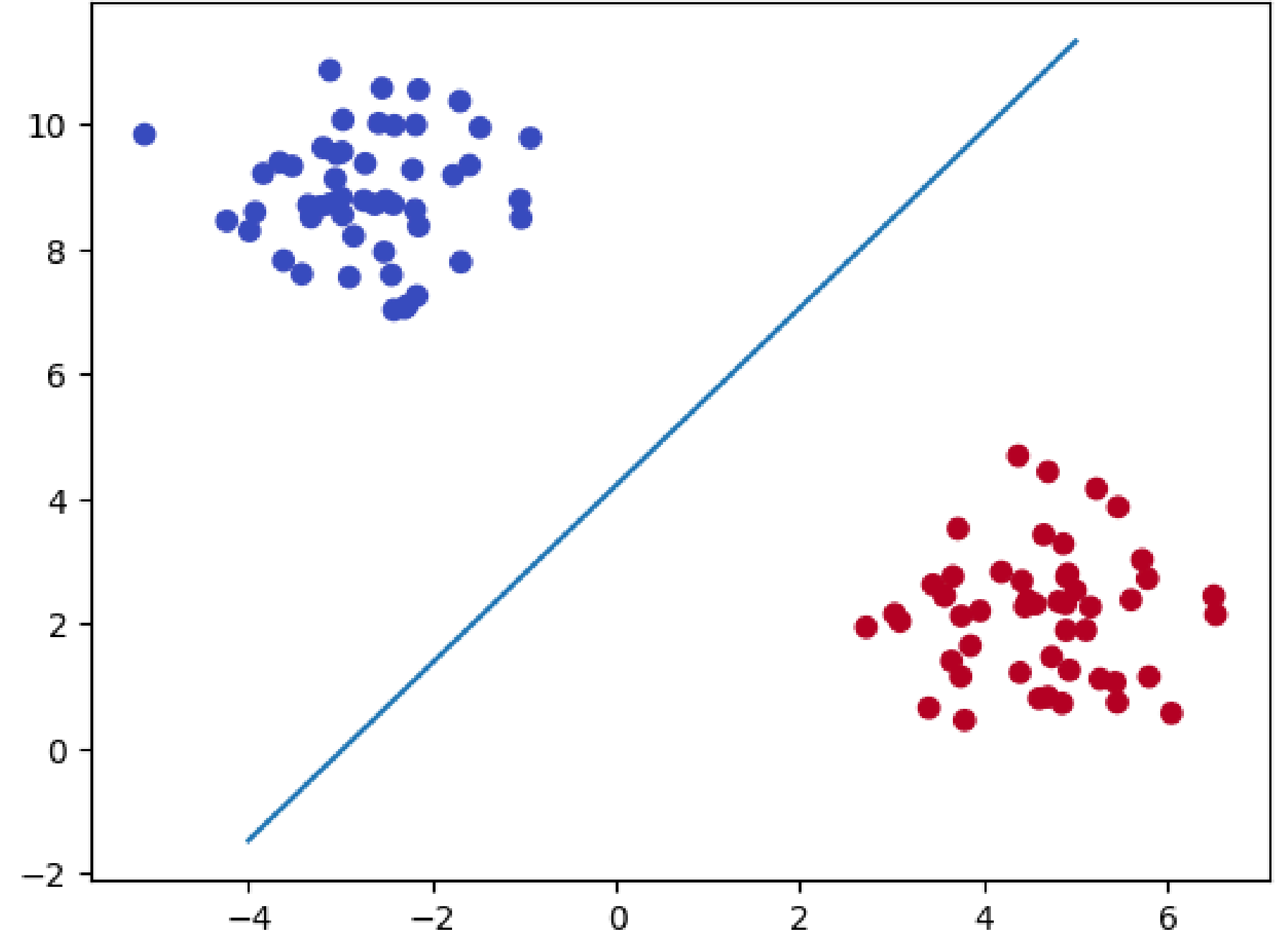
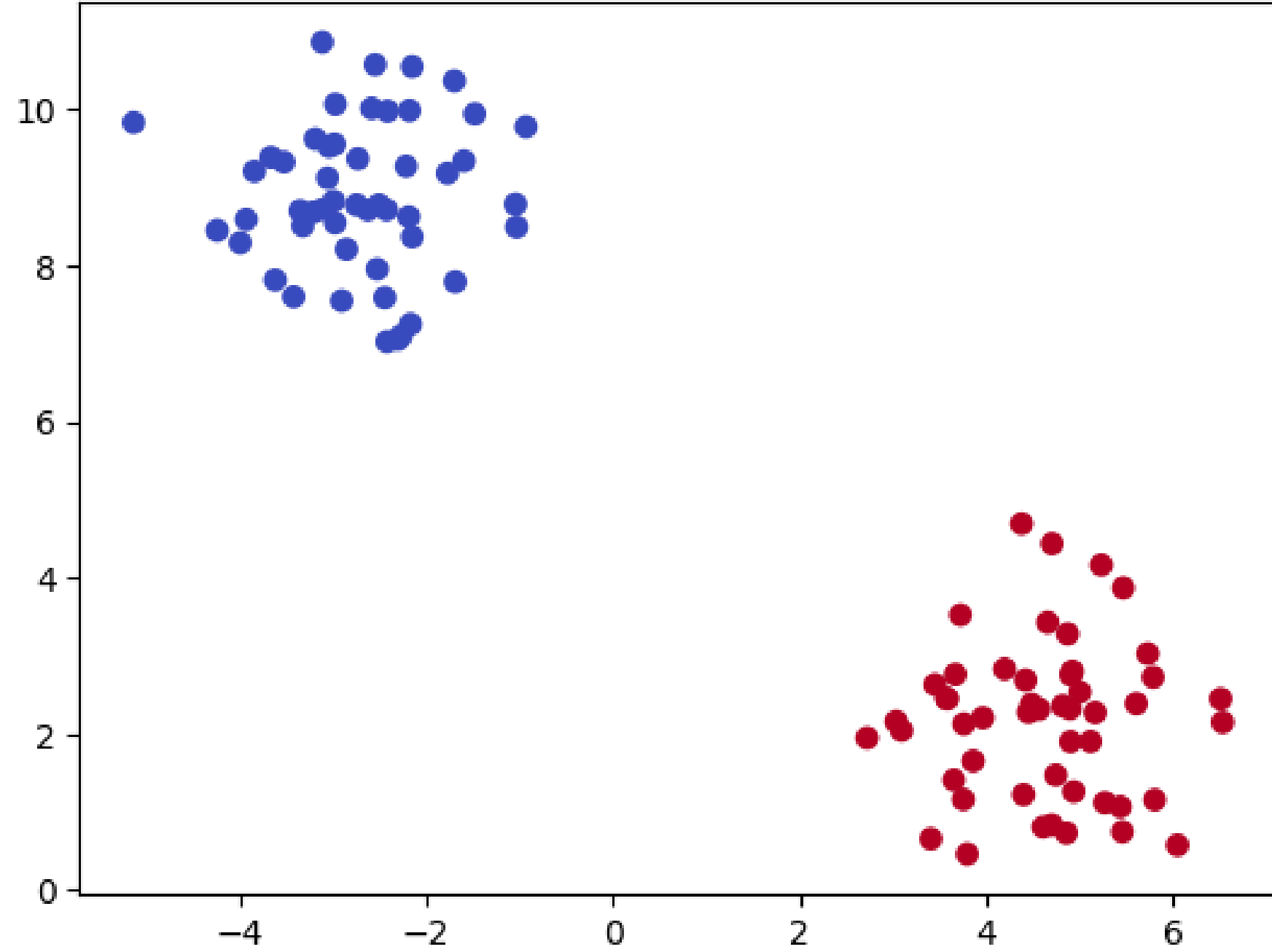
05 - KERNEL TRICK

06 - EXAMPLE

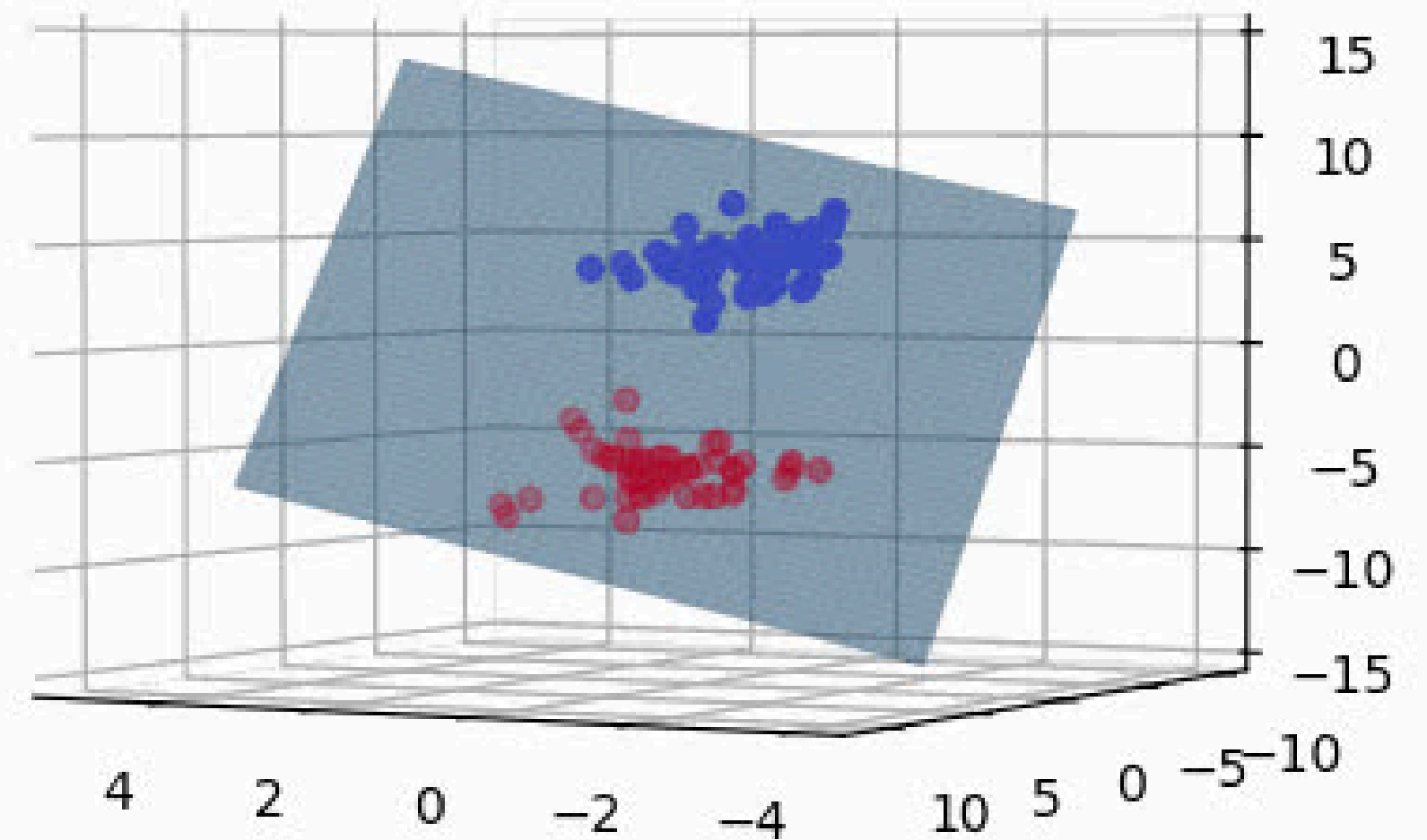
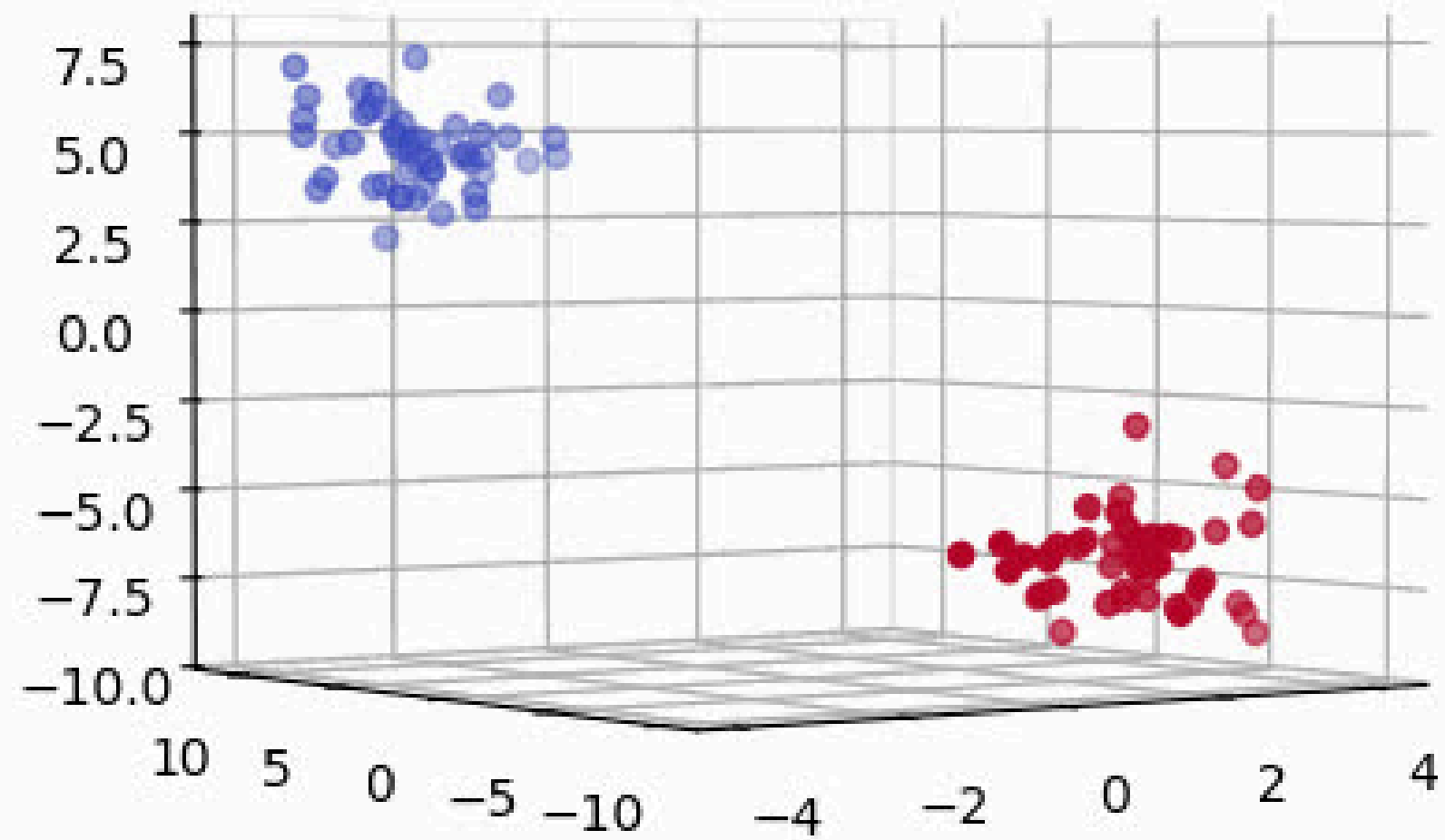
PROJECT TIMELINE



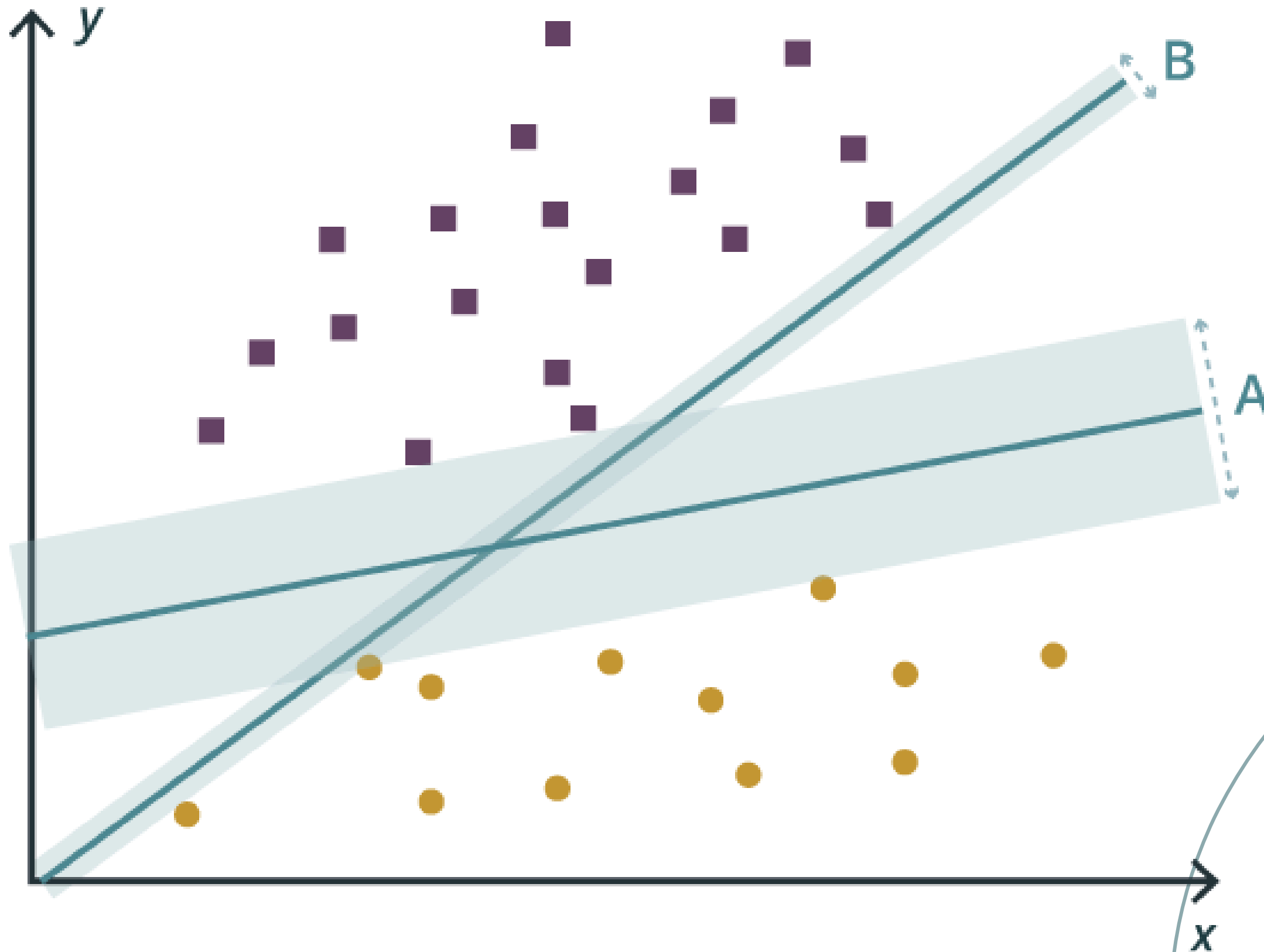
Hyperplane (2d) line



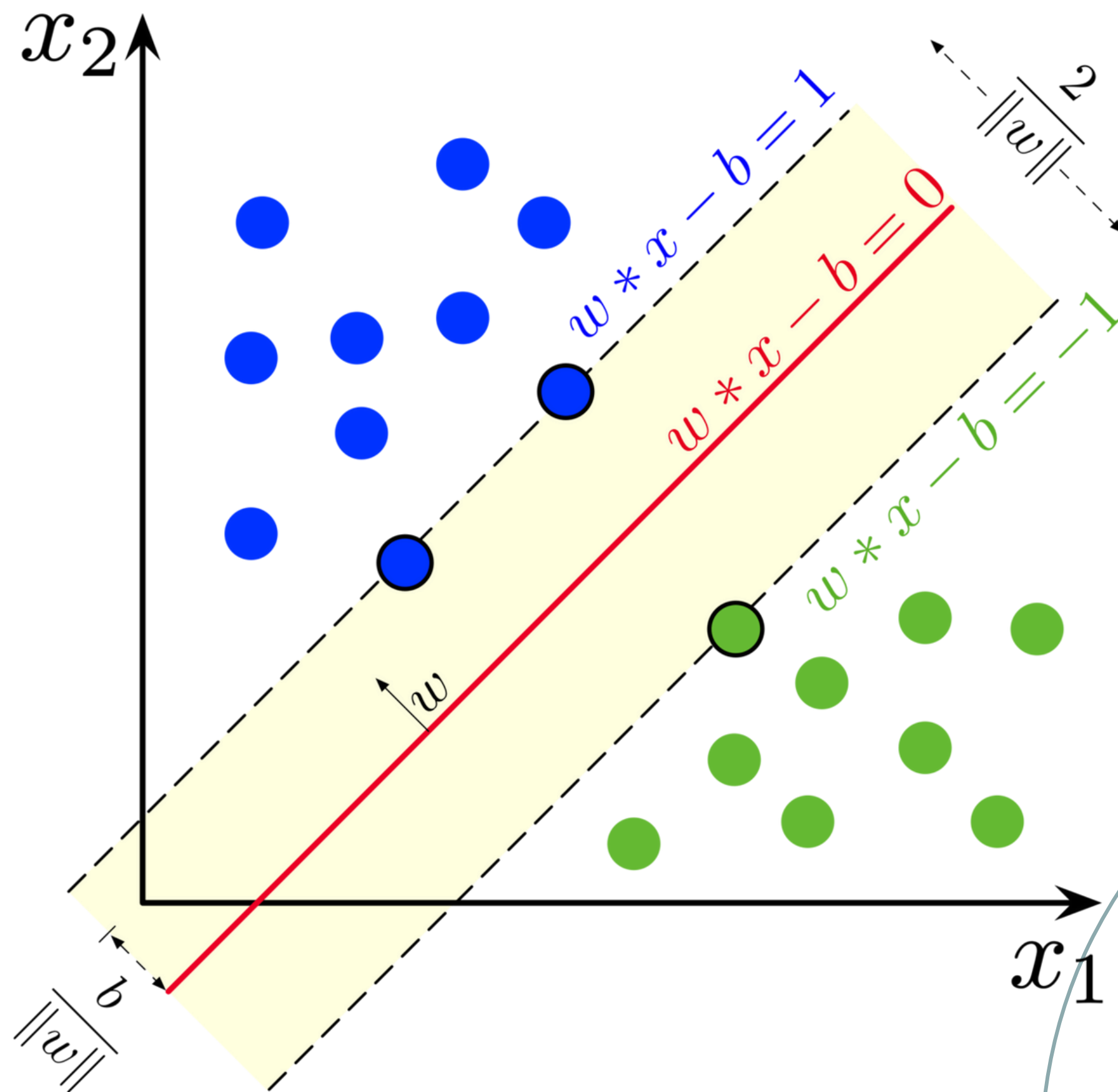
Hyperplane (3d) : plane



how to find the best hyper plane ?



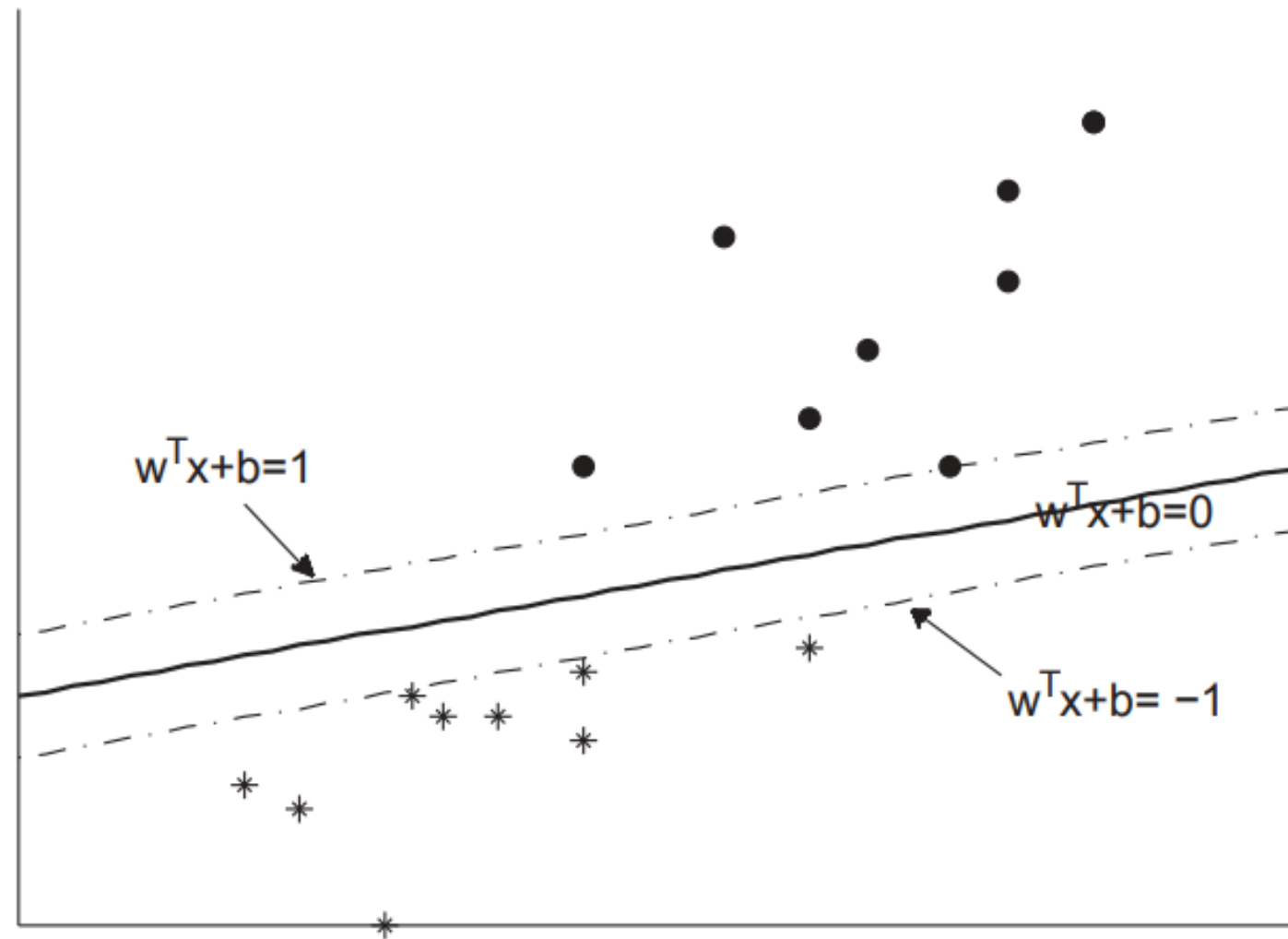
SVM PRINCIPLE



Optimization problem

We are looking for a hyperplan such as :

- The distance between the closest point and the plane is maximal.
- The points are still in the good side of the plane.



$$d = 2r = \frac{2}{||w||}.$$

$$y_i(w^T x_i + b) \geq 1$$

Optimization with constraints

$$\min \frac{1}{2} ||w||^2 \quad s.t. \quad y_i(w^T x_i + b)$$

$$\rightarrow L(w, b, \alpha) = \frac{1}{2} ||w||^2 - \sum_{i=1}^n \alpha_i [y_i(w^T x_i + b)]$$

Quadratic Problem..

$$\frac{\partial L(w, b, \alpha)}{\partial w} = 0 \quad \Rightarrow \quad w = \sum_{i=1}^n \alpha_i y_i x_i,$$

$$\frac{\partial L(w, b, \alpha)}{\partial b} = 0 \quad \Rightarrow \quad \sum_{i=1}^n \alpha_i y_i = 0.$$

dual coefficient

Optimization with constraints

maximise
$$\sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j (\mathbf{x}_i \cdot \mathbf{x}_j)$$

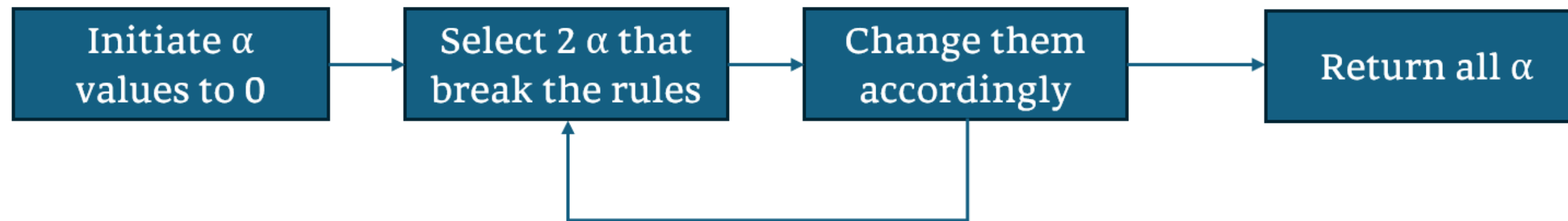
$$\sum_{i=1}^N \alpha_i y_i = 0$$

dual coefficient

How ?

Optimization with constraints

Using Sequential minimal optimisation



$$\alpha_a y_a + \alpha_b y_b = - \sum_{N \setminus \{a,b\}} \alpha_i y_i$$

condition to respect (Karush-Kuhn-Tucker (KKT))

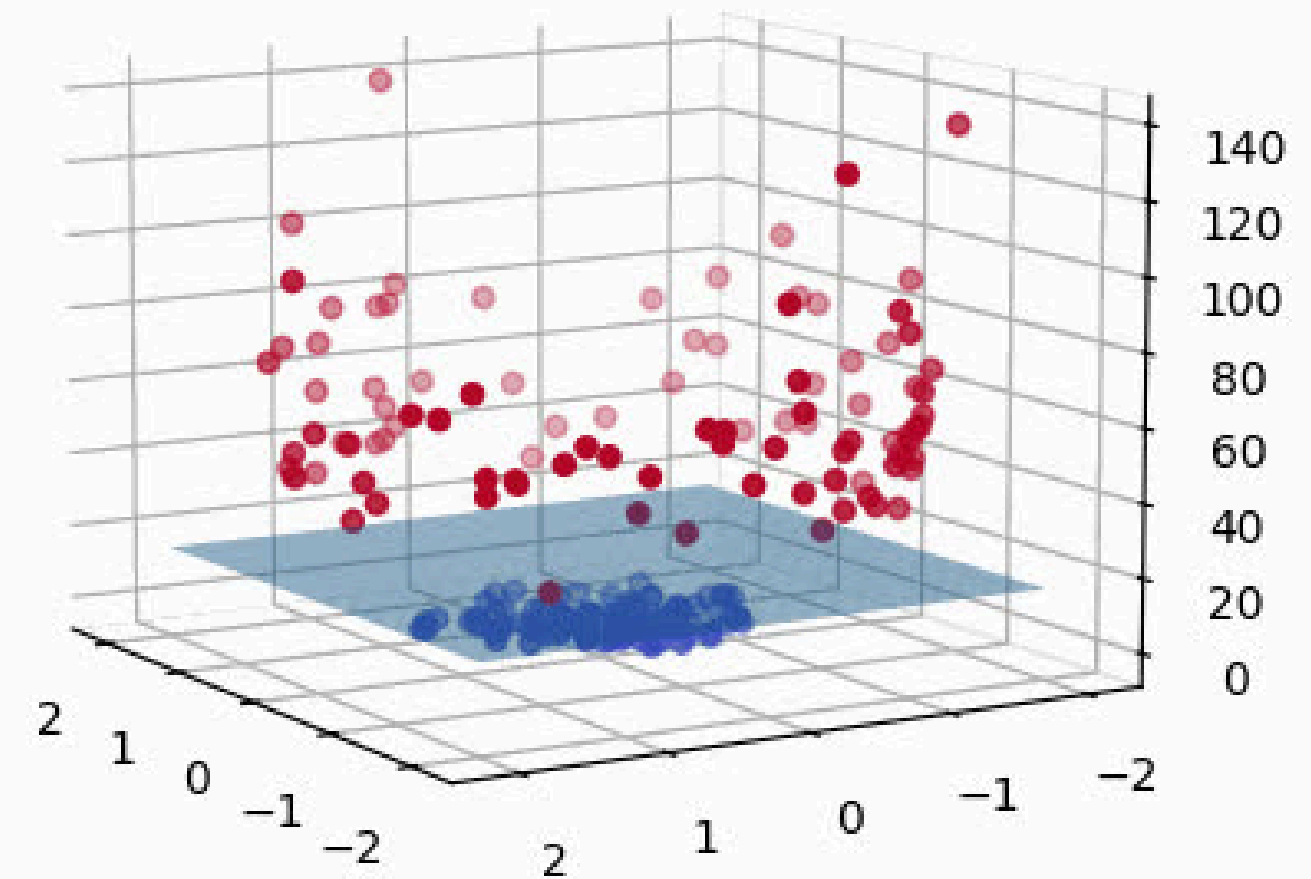
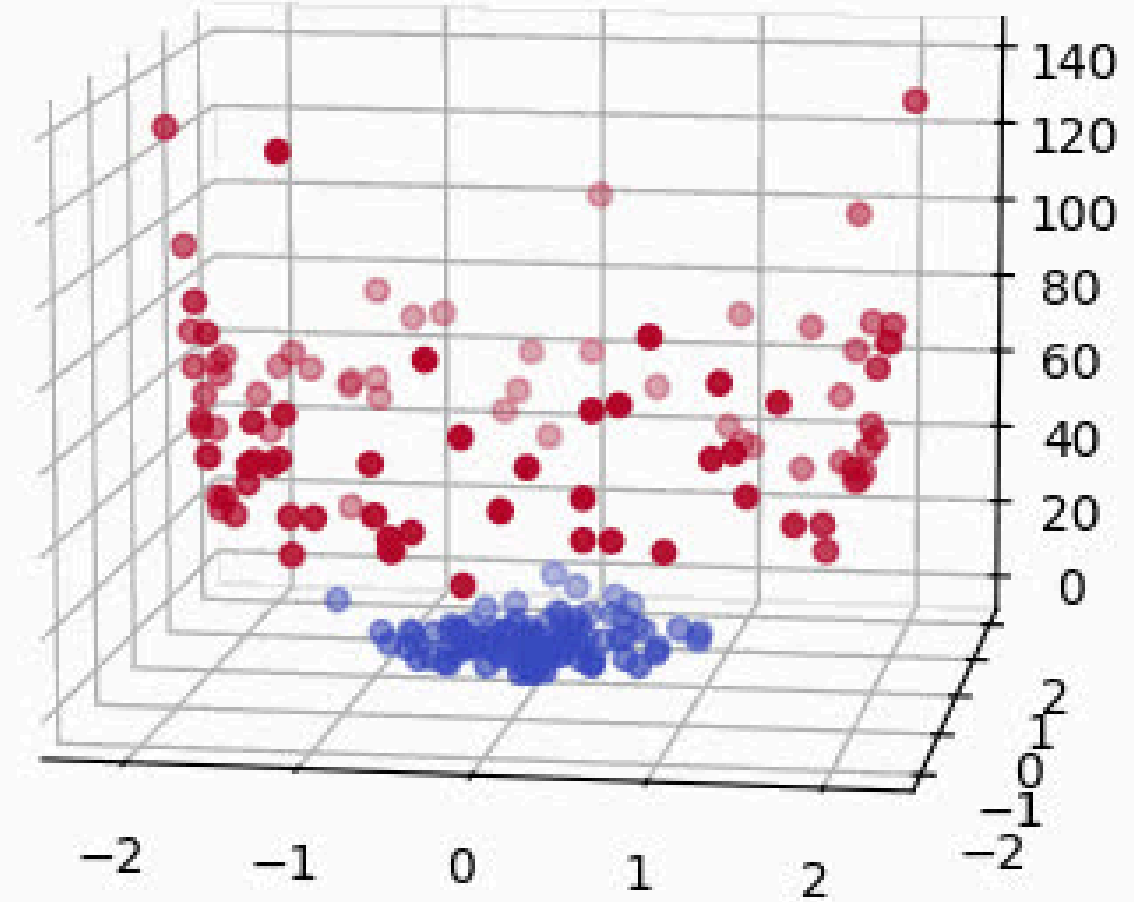
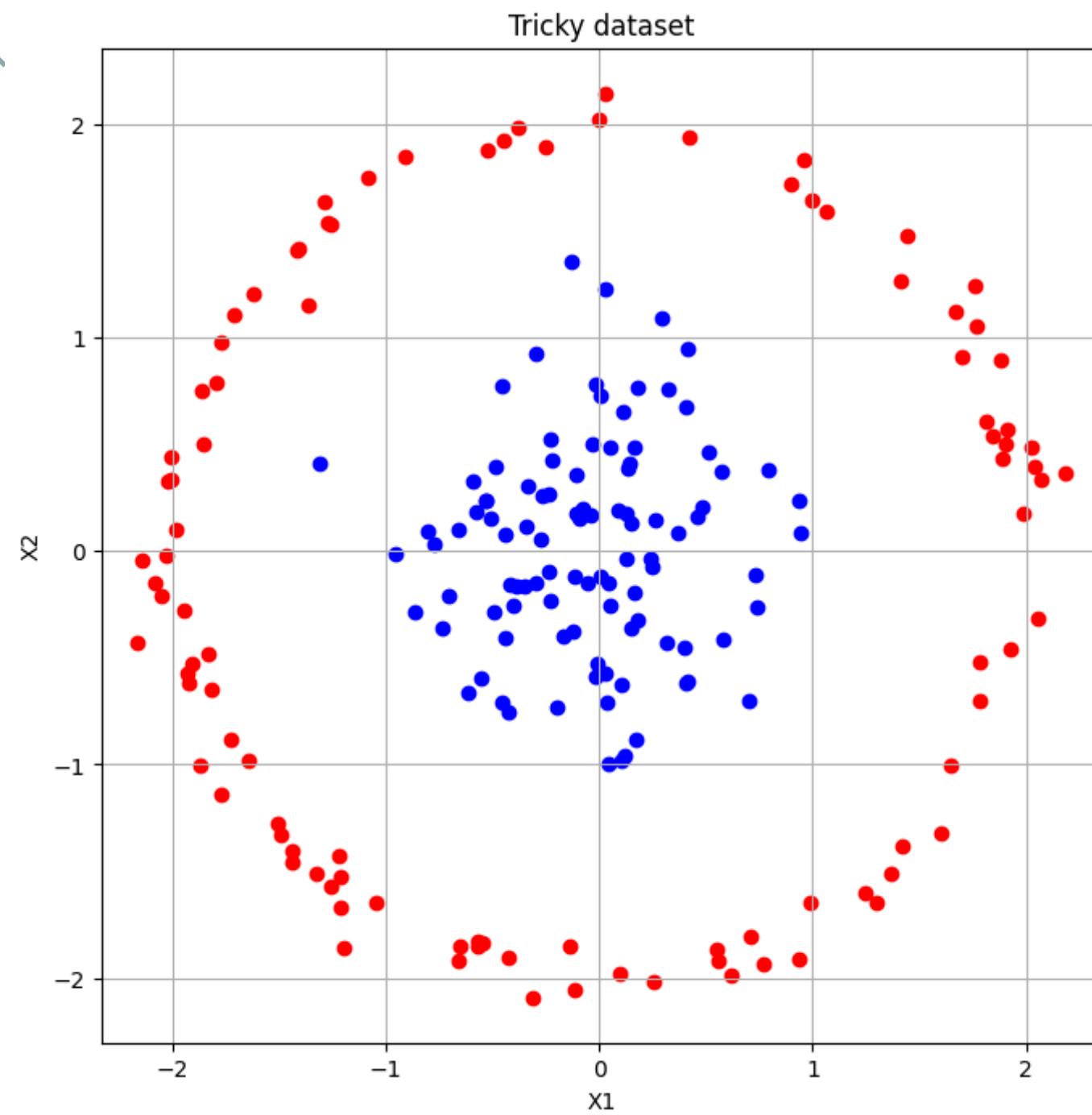
$$\begin{aligned} w \cdot x_i + b &\geq 1 \text{ if } y_i = 1 \\ w \cdot x_i + b &\leq -1 \text{ if } y_i = -1 \\ 0 &\leq \alpha_i \leq C \end{aligned} \quad \sum_{i=1}^N \alpha_i y_i = 0$$

Optimization with constraints

$$w = \sum_{i=1}^n \alpha_i y_i x_i, \quad \bar{b} = -\frac{1}{2} \bar{w} [X_r + X_s].$$

With X_r and X_s being the support vector of each categories

Kernel Trick



Example of an SVM application

```
from sklearn.svm import SVC
import numpy as np
import matplotlib.pyplot as plt
```

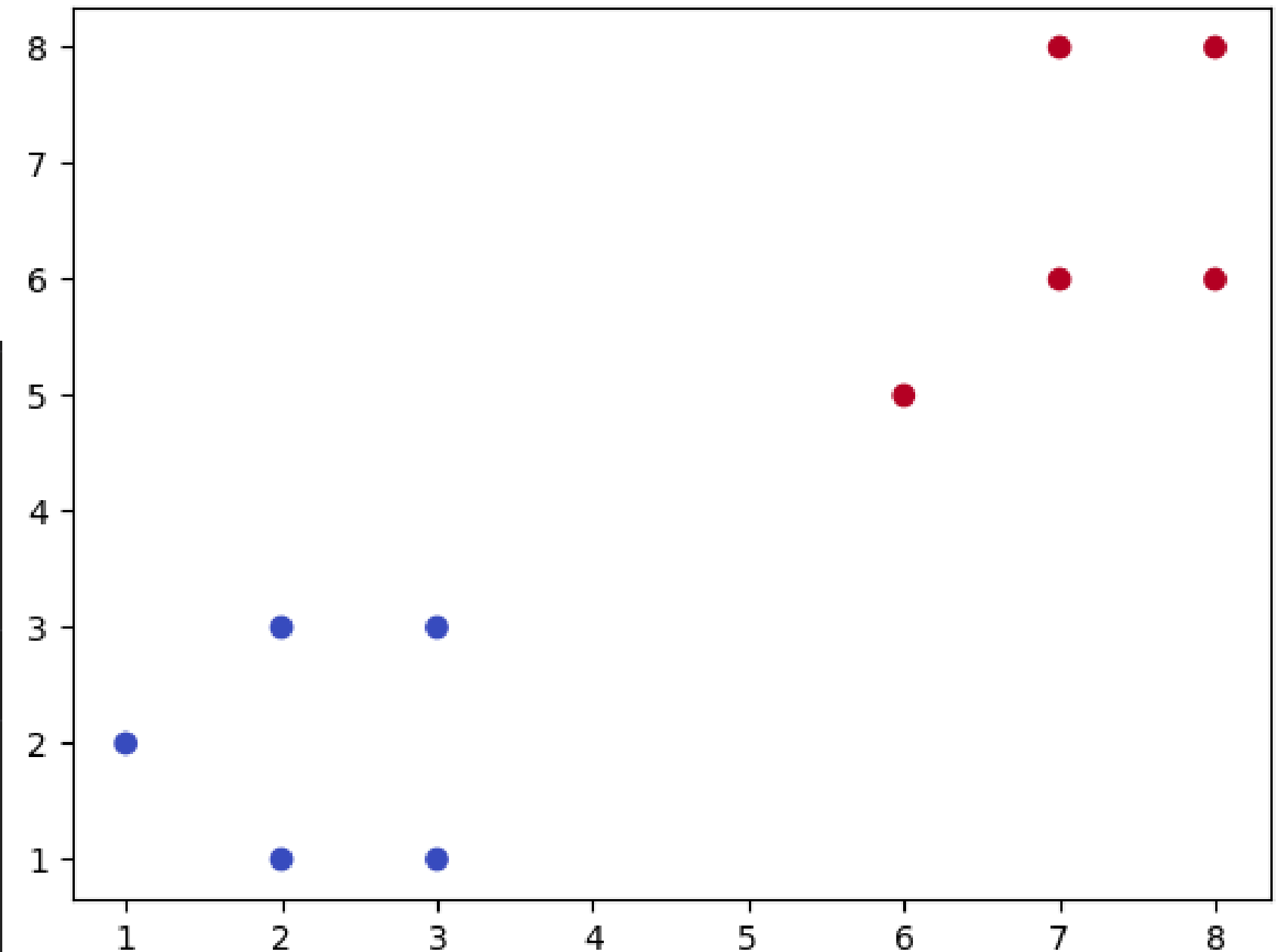
✓ 0.0s

```
# Sample dataset with 10 values (5 per class)
X = np.array([
    [1, 2], [2, 3], [3, 3], [2, 1], [3, 1], # Class -1
    [6, 5], [7, 8], [8, 8], [7, 6], [8, 6] # Class +1
])
y = np.array([-1, -1, -1, -1, -1, 1, 1, 1, 1, 1])
```

✓ 0.0s

```
plt.scatter(X[:, 0], X[:, 1], c=y, cmap='coolwarm')
```

✓ 0.0s



Example of an SVM application

```
data = pd.DataFrame(X, columns=['X1', 'X2'])
data['y'] = y
data['dual'] = [0 for _ in range(len(data['X1']))]
data.loc[svm.support_, 'dual'] = dual_coefs[0]
data
```

✓ 0.0s  Open 'data' in Data Wrangler

```
# Train SVM model
svm = SVC(kernel="linear", C=1)
svm.fit(X, y)

# Retrieve support vectors and alphas
support_vectors = svm.support_vectors_
dual_coefs = svm.dual_coef_ # This is the product of y_i * alpha_i
```

```
# Output the results
support_vectors, dual_coefs[0]
```

✓ 0.0s

```
(array([[3., 3.],
        [6., 5.]]),
 array([-0.15384615, 0.15384615]))
```

$$\overline{w} = \sum_{i=1}^n \overline{\alpha}_i y_i x_i,$$

X1	X2	y	dual
1	2	-1	0.000000
2	3	-1	0.000000
3	3	-1	-0.153846
2	1	-1	0.000000
3	1	-1	0.000000
6	5	1	0.153846
7	8	1	0.000000
8	8	1	0.000000
7	6	1	0.000000
8	6	1	0.000000

Example of an SVM application

$$w^T x + b = 0,$$

```
w = [0, 0]
for i in range(len(support_vectors)):
    w += dual_coefs[0][i] * support_vectors[i]
w
```

✓ 0.0s  Open 'w' in Data Wrangler

array([0.46153846, 0.30769231])

```
b = -1/2 * sum((np.dot(w, x) + b) for x in support_vectors)
b
```

✓ 0.0s

-3.3076923076923075

$$\bar{b} = -\frac{1}{2}\bar{w}[X_r + X_s].$$

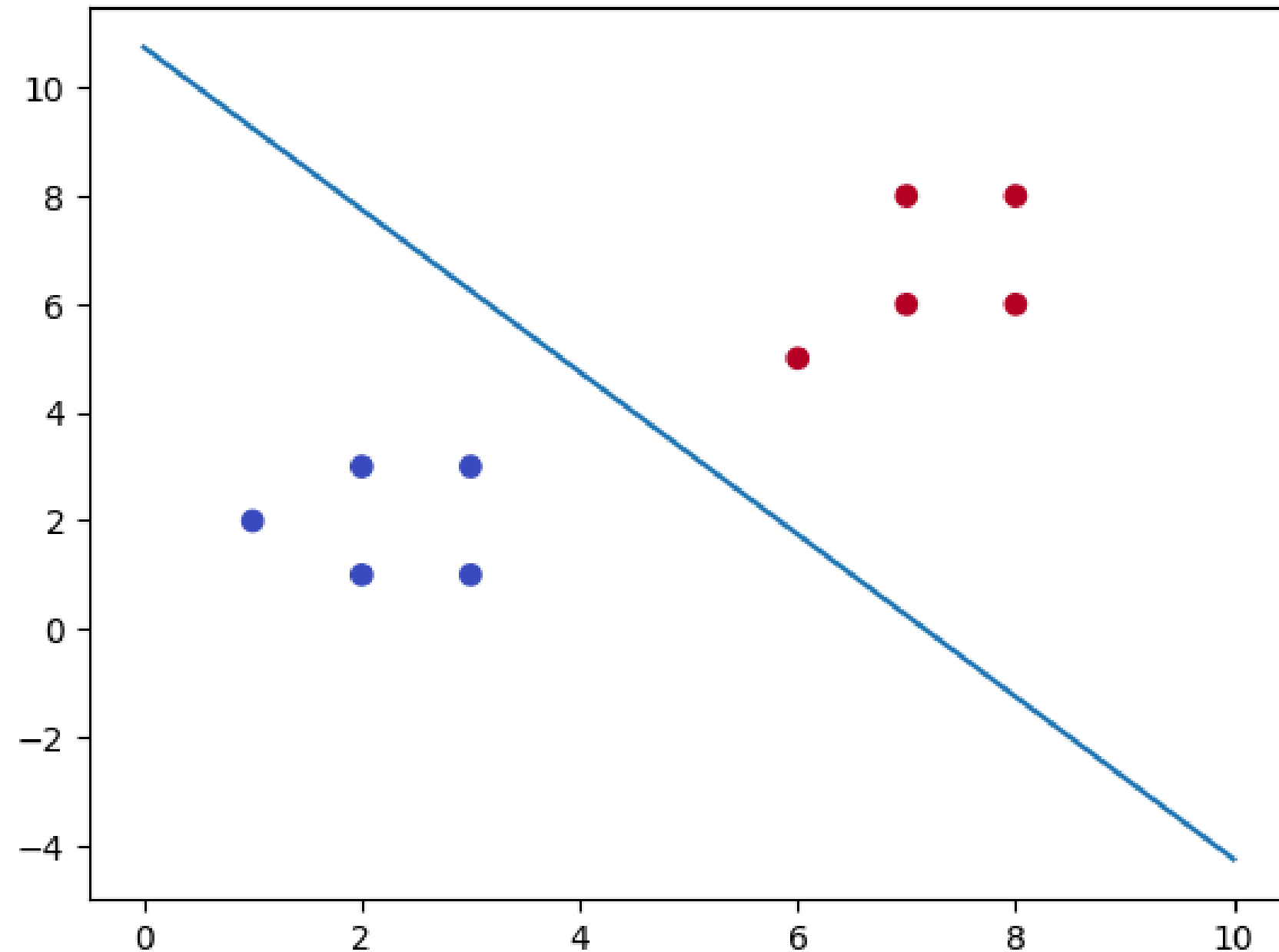
$$\bar{w} = \sum_{i=1}^n \bar{\alpha}_i y_i x_i,$$

Example of an SVM application

```
print(f'{round(w[0],2)} * x1 + {round(w[1],2)} * x2 + ({round(b,2)}) = 0')  
print(f'x2 = {round(-w[0]/w[1],2)} * x1 + {round(-b/w[1],2)}')
```

✓ 0.0s

```
0.46 * x1 + 0.31 * x2 + (-3.31) = 0  
x2 = -1.5 * x1 + 10.75
```





**THANK YOU FOR
LISTENING !
*FEEL FREE TO ASK ANY
QUESTION !***

REFERENCES :

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Research Survey on Support Vector Machine

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